

Extended Abstract

Motivation Sepsis is a life-threatening syndrome triggered by the body’s extreme response to infection, and it poses a significant clinical challenge due to its heterogeneous causes and the need for continuous, individualized interventions such as IV fluids and vasopressors. While reinforcement learning (RL) has demonstrated promise for optimizing sequential treatment decisions under uncertainty, the “black box” nature of RL models limits clinical trust and adoption. In this work, we introduce a two-phase framework that combines predictive performance with interpretability. Our approach yields transparent, human-readable treatment logic that bridges the gap between algorithmic intelligence and actionable clinical insights.

Method We leverage the publicly-available MIMIC-III database — comprising more than 53,000 adult ICU admissions with rich data on demographics, laboratory test results, “chart events” (recording patient state and vital signs), and “input events” (recording treatments and interventions) — to construct a sepsis cohort of over 17,000 patients. We further evaluate model performance on a smaller subset that includes simulated data designed to mirror low-resourced settings. In Phase I, we train a deep Q-network (DQN) on sepsis trajectories and enhance robustness by ensembling multiple independent agents and experimenting with behavior cloning (BC) regularization. In Phase II, we distill the learned policies into shallow decision trees via greedy, probabilistic, and Q-value methods.

Implementation We model our state space as a combination of patient clinical features such as heart rate, oxygen saturation, and mean arterial pressure, and our action space as discrete combinations of IV and vasopressor levels. Our reward penalizes in-hospital deaths and aggressive interventions while rewarding long-term survival. First, we train a baseline DQN model that utilizes the standard components of a replay buffer and a separate target network for stability. We note that deep Q-learning is well-suited to such a high-dimensional patient state with discrete actions. For ensembling, we train multiple DQNs and average their value estimates into a consensus policy. For regularization, we add a supervised BC term that penalizes deviations from clinician actions, with a tunable weight that lets us balance between pure RL and pure imitation. We use three approaches for distilling shallow decision trees from these learned policies: (1) in greedy distillation, we train a classification tree directly on the highest-value actions; (2) in probabilistic distillation, we convert the value estimates into action-probability distributions and train a classification tree on those soft targets; and (3) in Q-distillation, we train a regression tree to predict the full action-value vector.

Results & Discussion The baseline DQN and ensemble DQN models do not demonstrate particularly high clinician agreement (0.231 and 0.224, respectively). These results may be attributed to the RL models attempting to discover novel treatment trajectories (whereas clinicians cannot experiment with sepsis treatment without facing significant practical and ethical challenges), as well as the influence of a smaller dataset (which reflects resource scarcity in real-world applications). We note that the slight drop in accuracy after ensembling likely correlates with increased robustness. After adding regularization with supervised BC that imitates clinician demonstrations, performance improves significantly. We naturally achieve the best clinician agreement (0.910) when imitation is highly weighted, but we are able to achieve similarly high agreement (0.906) even with very modest regularization. The distilled decision trees adhere very well to the underlying learned policies. All three distillation methods achieve over 0.85 fidelity to the ensemble policy and over 0.95 fidelity to the BC-DQN policy. These results suggest that shallow decision tree surrogates can capture learned policy with minimal losses in alignment but significant improvements with interpretability.

Conclusion Our experiments show that deep Q-learning agents, when enhanced with ensemble methods and behavior-cloning regularization, can learn personalized sepsis treatment policies that reduce action-selection variance under noisy inputs and improve agreement with historical clinician decisions. Distilling these complex policies into shallow decision trees preserves essential structure while rendering decisions much more transparent. In future work, we are interested in exploring more sophisticated reward designs that may penalize intermediate adverse events (e.g., organ failure), investigating safe online RL frameworks for real-time adaptation under strict exploration constraints, and developing a clinician-facing interface to collect feedback and refine treatment policies via human-in-the-loop learning — bringing us closer to trustworthy and widely adopted AI support for critical-care decision making.

From Black Box to Bedside: Distilling Reinforcement Learning for Interpretable Sepsis Treatment

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Abstract

Sepsis is an extremely complex disease whose treatment requires constant monitoring and intervention. Previous reinforcement learning (RL) models have shown success in learning sepsis treatment plans, but they suffer from a lack of clinical trust. We present a two-phase approach that aims to enhance both the efficacy and interpretability of sepsis management models. We first train a baseline deep Q-learning model using a deep Q-network (DQN) and explore augmentation with ensembling and behavior cloning (BC) regularization. We then distill the complex learned policies into decision trees using a variety of approaches. We find that RL with modest imitation regularization incorporating clinician expertise and tree distillation with Q-values present a unique balance of performance, robustness, and transparency, which may enable more widespread adoption of medical AI.

1 Introduction & Motivation

Sepsis — a life-threatening disease caused by the body’s extreme response to an infection — remains a leading cause of death in ICUs worldwide. It is a uniquely difficult condition to treat because there are many possible underlying causes, including common infections; it is hard to diagnose; and it can worsen and lead to multi-organ failure very quickly. Sepsis care requires continuous monitoring and frequent administration of medications such as intravenous (IV) fluids, vasopressors, and antibiotics.

Previous research indicates that reinforcement learning (RL) has shown great promise in optimizing personalized sepsis treatment strategies. Indeed, sepsis management presents an ideal environment for RL due to its reliance on sequential decision making under high uncertainty. However, one major challenge is the lack of *interpretability* behind RL models which currently operate as “black boxes”, making it difficult for medical experts to trust and adopt these technologies and their corresponding medical recommendations.

We propose a novel two-phase framework that balances model performance and transparency. First, we utilize deep Q-learning and train a complex deep Q-network (DQN) on sepsis patient data from the publicly-available Medical Information Mart for Intensive Care (MIMIC-III) dataset developed by Johnson et al. (2016). We also apply ensemble methods and imitation regularization to improve robustness and clinician agreement. Second, we distill the learned policy into a more interpretable decision tree using a variety of techniques. We explore how these methods increase the understanding behind models’ decision-making and allow clinicians to thoroughly evaluate treatment logic. Our hybrid approach addresses a central challenge in medical AI: translating algorithmic intelligence into actionable, credible, and human-readable insights.

2 Related Work

Research from Komorowski et al. (2018) has indicated that RL policies trained on historical ICU data can outperform state-of-the-art methods that clinicians are currently using to treat sepsis. Tu et al. (2025) introduced RL policies for sepsis care, trained with conservative Q-learning, that aimed to improve model safety by discouraging rare or excessively extreme sepsis treatment recommendations. However, given the complex algorithms that these models rely on, there exists significantly less transparency and trust for adoption in the real world. In an attempt to improve model interpretability without sacrificing performance, many research groups have begun to adopt new approaches like policy distillation, where a simple model is trained to mimic the decisions of a more complex one. Pettit et al. (2021) introduced one method for learning sparse symbolic policies for sepsis treatment from an initial neural network-based policy, aiming to retain decision-making efficacy with the most simple possible resulting policy.

Bastani et al. (2019) argued that decision trees are a particularly promising class of interpretable models because they are nonparametric yet highly structured; as such, they are a good candidate to represent the complex policies learned by neural networks. Soft decision trees combine the learnability of neural networks with the interpretability of simpler models by using probabilistic (soft) instead of hard splits, as studied by Coppens et al. (2019). Recent work from Kohler et al. (2025) further explored the distillation of deep neural networks into decision trees and multi-layer perceptrons and showed that more interpretable models can maintain expert performance. All of these approaches facilitate the translation of complex RL policies into formats that are more accessible to clinicians, potentially improving trust and accessibility.

Additional studies emphasize the importance of human-centered design in clinical settings. One insight to explore is the integration of human expertise into RL models. Che et al. (2017) wrote about the importance of interpretable clinical decision support systems: they stressed that transparency is critical for widespread adoption, especially in high-stakes environments like ICUs. Similarly, Wu et al. (2023) proposed a value-based deep RL model that incorporates human expertise to enhance the safety and reliability of sepsis treatment recommendations. Such work underscores the ongoing need to complement model development — even after distillation — with clinical expertise in order to create truly trustworthy AI systems.

3 Method

3.1 Reinforcement Learning

For the first phase, we used deep Q-learning for our base reinforcement learning algorithm. The (basic) Q-learning algorithm approximates a Q-function

$$Q^\pi(s, a) = \sum_{t'=t}^T \mathbb{E}_\pi[r(s_{t'}, a_{t'}) \mid s_t, a_t]$$

representing the expected future rewards from starting at state s_t , taking action a_t , and then following policy π . In particular, it learns the optimal Q-function Q^* by iteratively updating its estimate as

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right),$$

where α is the learning rate, r is the immediate reward from taking action a , γ is the discount factor, and $\max_{a'} Q(s', a')$ is the greedy estimate of the future return from state s' . In basic Q-learning, the Q-values for each state-action pair are stored in a table, but this becomes impractical for high-dimensional state spaces. Deep Q-learning, instead, trains a neural network (a DQN parametrized by θ) to approximate the Q-values for all actions from a given state s . It does so by minimizing the loss

$$\mathcal{L}(\theta_i) = \mathbb{E}_{s, a, r, s'} \left[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta_i))^2 \right].$$

Note that transitions (s, a, r, s') are sampled from the replay buffer, and the target network θ^- is periodically updated to improve stability.

We chose to use deep Q-learning for several reasons:

- It is particularly well-suited for a high-dimensional state space alongside a discrete, low-dimensional action space. (We expand more on our experimental design in the next section.)
- It learns an explicit Q-function, which provides a useful method for distillation in our second phase that preserves some notion of “valuable” treatment options.
- It is off-policy, allowing it to learn the *optimal* policy regardless of whether the data was collected under an exploratory *behavior* policy — which is typically not the case in medical settings, since clinicians cannot ethically or practically experiment with treatment options on real sepsis patients.

We explored two variations on our baseline model. We first trained an ensemble of DQNs with different seeds and averaged Q-values across models to produce a more robust policy.

We also introduced behavior cloning (BC) regularization. Our hybrid objective combined RL with supervised BC, using the clinician treatment actions from MIMIC-III as expert demonstrations:

$$\mathcal{L} = \mathcal{L}_Q + \alpha \cdot \mathcal{L}_{BC}.$$

This helped encourage policy alignment with clinicians while still preserving some insights from Q-learning. We experimented with hyperparameter tuning on different values of α to investigate the balance between imitation and exploration. (If α was 0, we recovered vanilla deep Q-learning; if α was very large, we could potentially overfit to the demonstrations and lose the benefits of RL.)

3.2 Decision Tree Distillation

For the second phase, our emphasis was on interpretability. We tried several approaches for distilling the ensemble policy and the BC-regularized policy into decision trees. We enforced a maximum depth of 5 to 6 levels: while decision trees as a class of models are more interpretable than neural networks, they can grow so deep that they essentially reconstruct much of the original network’s complexity, which negates the original gains in interpretability.

Our first approach was greedy distillation. We trained a classification tree whose target label at each state s_i was the ensemble’s best action $y_i = \arg \max_a Q(s_i, a)$. The optimal splits were calculated by minimizing the Gini impurity, which measures the likelihood of misclassifying new data (i.e., of choosing the wrong action at a given state). If we consider a dataset D at a node containing samples from k different classes, and the probability of samples belonging to a particular class i is p_i , then the Gini impurity at that node is

$$G(D) = 1 - \sum_{i=1}^k p_i^2.$$

Our second approach was probabilistic distillation. Instead of using the raw Q -values to select target labels, we took the softmax of the values at state s_i , which yielded a probability distribution over the possible actions. We randomly sampled an action from this distribution to obtain a target label and trained a classification tree using the same splitting criterion (Gini impurity).

Our third approach was “Q-distillation”. In this approach, we trained a regression tree instead of a classification tree, treating the Q -values for all actions at state s_i as a “Q-vector”:

$$\vec{q} = (Q(s_i, a_1), Q(s_i, a_2), \dots, Q(s_i, a_{10})).$$

The optimal splits were calculated by minimizing the mean squared error (MSE). If we consider a dataset D at a node with Q-vectors $\vec{q}_i$ and mean Q-vector $\vec{\mu}$, then the MSE at that node is

$$MSE(D) = \frac{1}{|D|} \sum_i \|\vec{q}_i - \vec{\mu}\|^2.$$

Splitting based on a particular feature partitions the data at a node into two child nodes D_L and D_R . We choose our best split to minimize the post-split weighted MSE

$$\frac{|D_L|}{|D|} MSE(D_L) + \frac{|D_R|}{|D|} MSE(D_R).$$

We used a variety of distillation approaches to further explore interpretability in medical settings and to understand the balance between learned insights and model transparency.

4 Experimental Setup

4.1 Dataset

We utilized data from the MIMIC-III dataset, a large, publicly-available database of de-identified patient data, developed by Johnson et al. (2016) from the Laboratory for Computational Physiology at MIT. MIMIC-III contains data on over 53,000 adult patients admitted to critical care units between 2001 and 2012; it includes important features like patient demographics, lab test results, “chart events” (including vital signs such as heart rate and oxygen saturation), and “input events” tracking the administration of medication or other interventions (such as IV fluids or vasopressors). A sepsis cohort of over 17,000 patients was extracted based on the International Classification of Diseases (ICD-9) codes recorded in the data.

To simulate low-resourced hospital environments and to explore several methods of distillation efficiently, we also used a subset of simulated clinical time series data derived from MIMIC-III by Khan (2022). This was particularly well-suited to the broader implications of our work and our goal of assessing model robustness and interpretability in settings that reflect real-world constraints. Yet another challenging component of sepsis management is ensuring reliable decision-making when faced with limited monitoring frequency, data availability, or previous clinical expertise. These conditions make it especially important to evaluate how well distilled models perform in low-data settings — where interpretability and resilience to noise become especially critical.

4.2 Experiment

We modeled sepsis patient care as a Markov decision process.

- The state space was represented by 24 clinical features (aforementioned chart events such as heart rate, oxygen saturation, mean arterial pressure, etc) extracted hourly from the ICU patient records.
- The action space was represented by 10 discrete treatment choices: combinations of 5 levels of IV fluids (0–4) and 2 levels of vasopressors (0, 1). These treatment actions were chosen because they are critical when patients enter septic shock and because they require constant monitoring and adjustments based on patients’ evolving needs. (In contrast, treatment actions such as administering antibiotics are typically one-time interventions.)
- The reward was initially designed with +15 for 90-day patient survival, −15 for in-hospital patient death, −1 for aggressive interventions (where the highest levels of both treatment actions were combined), and +0.01 per timestep that the patient remained alive.

We used the `Stable-Baselines3` library to train our baseline RL model for easier experimentation. For the reinforcement learning phase, we evaluated our models based on agreement with clinician decisions. For the distillation phase, we evaluated our decision trees based on their fidelity to the learned policies — i.e., the percentage of times where the actions predicted from the trees aligned with the learned policies.

5 Results

5.1 Reinforcement Learning

We measured the agreement between each of our RL models with the clinician actions.

Method	Clinician Agreement
Baseline DQN	0.231
Ensemble DQN	0.224
BC-Regularized DQN (best)	0.910
BC-Regularized DQN (low α)	0.906

Table 1: Agreement of RL models with clinician-labeled actions.

The vanilla DQN model, trained without any form of regularization or ensembling, achieved relatively low agreement (0.231) with clinicians. This indicates that its learned policy diverged substantially from standard treatment decisions. One possible explanation is that the baseline agent was attempting to discover novel treatment trajectories. An RL agent can experiment with different actions, whereas clinicians cannot experiment with sepsis treatment without facing significant practical and ethical challenges. However, low alignment with clinician expertise presents a concern for real-world adoption. Another factor contributing to this discrepancy could be the simulated underresourced environment that we used, with smaller datasets compared to the entirety of MIMIC-III: in clinical practice, densely sampled ICU data may not always be available. We observe that a pure RL model trained under these conditions risks recommending interventions that are clinically unacceptable without further safety checks.

The ensemble DQN model performed slightly worse than baseline DQN in terms of clinician agreement (0.224). However, averaging the Q-values during ensembling naturally reduced variance in action selection: even if one DQN’s estimate was overly optimistic or pessimistic, another network in the ensemble could err in the opposite direction and stabilize the overall Q-value estimates. These results demonstrate the long-recognized tradeoff between robustness and accuracy.

The BC-regularized DQN models performed significantly better. We achieved the highest clinician agreement of 0.910 using $\alpha = 1.0$, which strongly weighted the behavior cloning component that imitated clinician actions. In order to confirm that this was not a result of overfitting, we conducted extensive hyperparameter tuning on α . Even with a very small value such as $\alpha = 0.01$, the regularized model maintained strong agreement (0.906). This indicates that modest imitation regularization can still greatly enhance alignment without fully overriding the baseline RL objective.

5.2 Decision Tree Distillation

All three distillation methods demonstrated relatively high fidelity to the ensemble DQN policy.

Distillation Method	Fidelity to Ensemble DQN
Greedy Tree Distillation	0.999
Probabilistic Tree Distillation	0.887
Q-Value Tree Distillation	0.914

Table 2: Fidelity of decision trees in replicating ensemble DQN policy.

Although the greedy distillation method excelled at matching the original learned policy, we believe that Q-distillation would be the best for real-world adoption. Building a decision tree using a regression approach (as opposed to a classification approach) provides us with a much richer signal about the relative benefits of actions instead of a single “best” action. Such nuance is vital in medical decision-making and furthers our goal of distilling complex insights into interpretable models.

All three methods also performed extremely well when matching the BC-regularized policy.

Distillation Method	Fidelity to Ensemble DQN
Greedy Tree Distillation	1.000
Probabilistic Tree Distillation	0.998
Q-Value Tree Distillation	0.999

Table 3: Fidelity of decision trees in replicating BC-DQN policy.

In addition, they followed the same relative trend: the greedy approach performed (near) perfectly, Q-distillation exhibited a similarly high fidelity, and the probabilistic approach was the weakest. We hypothesize that probabilistic distillation underperforms because it uses a near-identical approach to greedy distillation, but sampling actions based on the distribution of softmax Q-values introduces additional randomness that degrades some benefits of greedy tree construction. Overall, such high fidelity to both policies suggests that decision tree distillation is a promising approach to improving interpretability — whether or not the underlying learned policy reflects high clinician agreement.

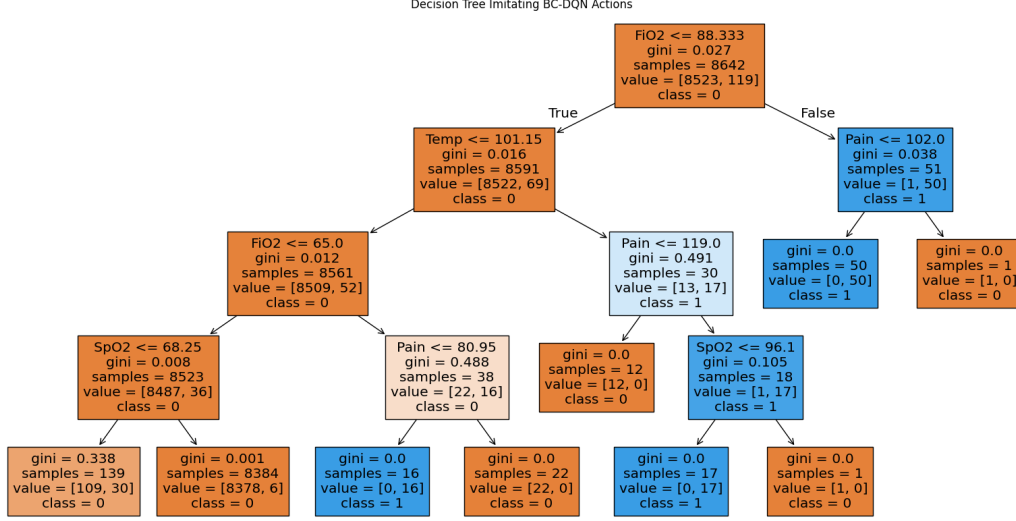


Figure 1: Decision tree built from BC-DQN using greedy distillation approach.

In Figure 1, we present an example of a decision tree distilled from the BC-DQN policy and note some key observations. First, most predicted actions (i.e., the class labels in the tree) are 0 or 1, indicating conservative treatment choices that align with real-world clinician decisions. Next, intuitive medical choices are already highlighted in greater clarity than from a black-box RL model. For instance, in the rightmost region of the initial left child, the tree recommends intervention if a patient is suffering from high pain and their oxygen saturation (SpO_2) falls below a certain threshold, while recommending no intervention for the time being if their SpO_2 levels are normal.

We also compare how *different* methods perform when distilling from BC-DQN. The underlying policy was trained with “Combo A” hyperparameters: $\alpha = 0.1$ (strongly de-emphasizing imitation) and learning rate 0.0001. An analysis of the three trees (Figures 2, 3, 4) is summarized in Table 4.

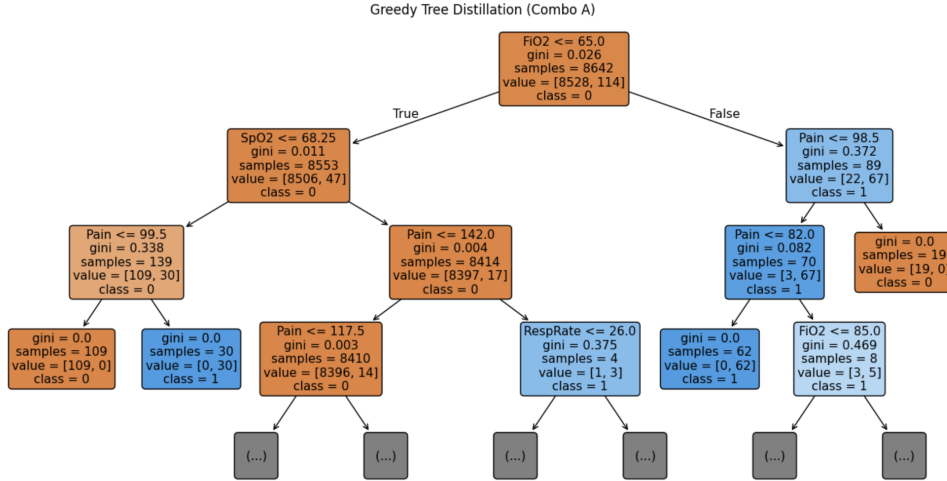


Figure 2: Greedy tree distillation (Combo A).

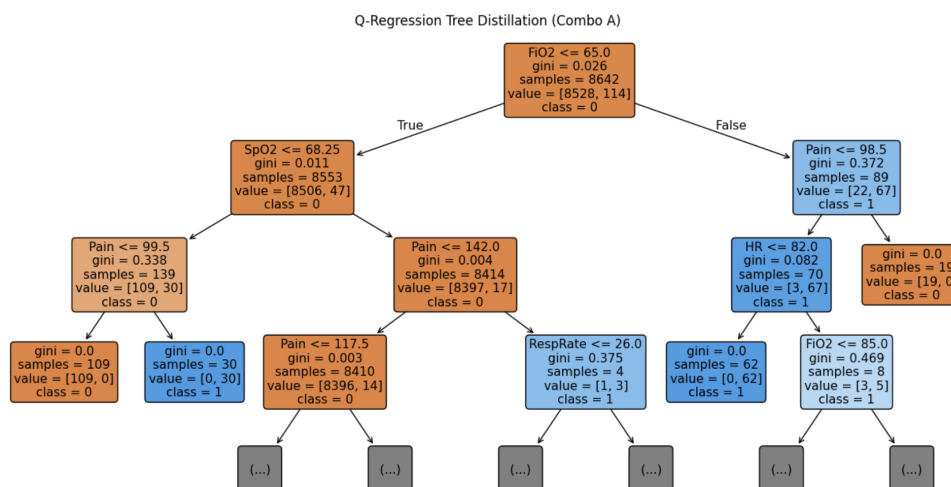
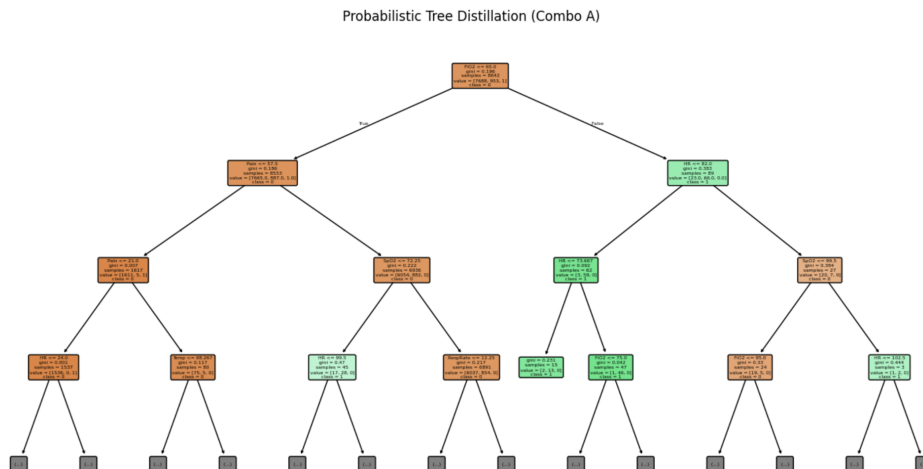


Figure 4: Q-regression tree distillation (Combo A).

Method	Split Criterion	Tree Depth	Notable Features
Greedy Tree	Gini impurity	Moderate (4–5)	Fastest computation; sharp thresholds; very interpretable
Probabilistic Tree	Gini impurity	Deeper (5–6)	Encodes more uncertainty; smoother decision boundaries; more complex but expressive
Q-Regression Tree	Q-value regression (MSE)	Moderate (4–5)	Optimizes for expected reward; continuous targets; aligns with value-maximizing

Table 4: Comparison of tree distillation methods (Combo A).

Finally, we note that our current work uses a moderately simple reward function that does not account for more detailed actions or patient outcomes, since we focused on two influential treatment options of IV fluids and vasopressors. Our baseline RL agents are also trained offline, and it may be worth exploring online RL using algorithms other than deep Q-learning. Further discussion of these limitations and future research directions are highlighted in our conclusion.

6 Conclusion

Our findings indicate that we can use deep reinforcement learning agents to effectively learn personalized sepsis treatment strategies from high-dimensional ICU data while simultaneously addressing the interpretability and robustness challenges that have hindered clinical adoption. Introducing ensembling methods and behavior cloning regularization on top of our baseline DQN model led to improvements in robustness and clinician agreement, respectively. In particular, BC regularization significantly enhanced performance, even with only minor influence from the imitation component.

We subsequently distilled the complex learned policies into shallow decision trees to ensure that outputs remained transparent and human-readable. Although greedy distillation achieved the highest fidelity overall, Q-distillation was nearly as successful at matching the learned policies while capturing more decision-making nuance.

Looking ahead, there is a rich array of both theoretical and practical avenues that we are interested in pursuing. One direction of interest is using more sophisticated reward functions that penalize intermediate unfavorable outcomes (such as organ failure or progression to a more severe state of sepsis) in addition to patient death, or that assign more fine-grained rewards for different treatment options, including a wider variety of treatment actions. Another direction is exploring online RL methods for sepsis management. This aligns more closely to real-world treatment, but also presents significant ethical and practical risks that may necessitate stringent safety constraints and conservative exploration strategies. A particularly crucial direction is integrating actual clinician feedback for human-centered design and development of medical AI. Ultimately, we believe that combining computational insight with human wisdom does not simply automate our decision-making — it elevates our collective capacity to heal.

Team Contributions

- **Andrea:** Preprocessed data from MIMIC-III; implemented reinforcement learning models and assisted with BC-regularized DQN.
- **Ella:** Implemented methods for interpretable decision tree distillation; suggested and led implementation of BC-regularized DQN.
- Both members contributed equally to the writing of the report.

Changes from Proposal We added a new type of model (BC-regularized DQN) and conducted hyperparameter tuning, because we noticed that our baseline and ensemble DQN models underperformed initial expectations. We also experimented with varying our dataset sizes to further explore real-world implications of resource availability, not only in terms of interpretability but also accessibility and robustness. We did not build out a UI prototype to collect clinician feedback because our primary focus was on refining the RL and distillation phases as described. However, we would be interested in continuing this research in the future to enable transparent, widely adoptable medical AI.

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